Z-TEST — Complete Study Guide

What is a Z-Test?

A Z-Test is a statistical method used to determine whether there is a significant difference between a **sample mean** and a **population mean**. It tells us — *"Is the difference we see just by chance, or is it real?"*

CONDITION 1	CONDITION 2
Population Standard Deviation (σ) is **KNOWN**	Sample Size **n > 30**

The Z-Test Formula

$$Z_d = (x■ - \mu) / (\sigma / \sqrt{n})$$

Symbol	Meaning	Example (Problem 1)
x■	Sample Mean (what you measured)	169.5 cm
μ	Population Mean (claimed value)	168 cm
σ	Population Std Deviation	3.9
n	Sample Size	36
Z_d	Calculated Z score	2.31

Decision Rules — When to Reject H0?

TEST TYPE	REJECT H0 IF...	p-VALUE RULE
Two-Tail (H1: μ ≠ value)	**Z < –1.96 OR Z > +1.96**	**p < 0.05**
Left-Tail (H1: μ < value)	**Z < –critical value**	**p < α**
Right-Tail (H1: μ > value)	**Z > +critical value**	**p < α**

H0 (Null Hypothesis) = The boring claim — "nothing changed, mean is still X"

H1 (Alternate Hypothesis) = What you want to prove — "mean has changed"

Alpha (α) = Significance level = 1 – Confidence Level   e.g. 95% CI → α = 0.05

PROBLEM 1 — City Height Example (Two-Tail Test)

A doctor believes the average height of city residents (μ **= 168 cm**) is different. He measures **36 individuals** and finds the average height to be **169.5 cm**. Population σ = 3.9. At **95% confidence level**, is there enough evidence to reject H0?

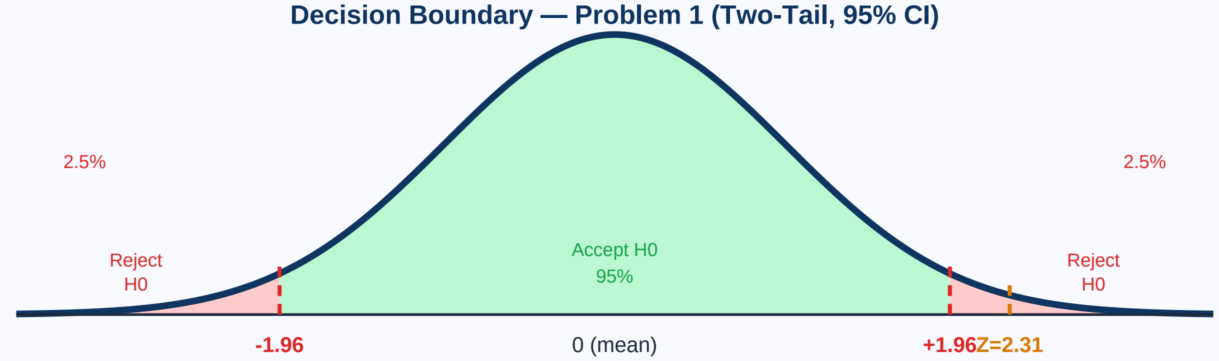
Given Values

μ (Population Mean)	σ (Std Dev)	n (Sample Size)	x■ (Sample Mean)	CI	α
168 cm	3.9	36	169.5 cm	95%	0.05

Step-by-Step Solution

1	**State Hypotheses**	H0: μ = 168 cm H1: μ ≠ 168 cm
2	**Find Critical Z Value**	CI = 95% → α From Z-table: a Reject H0 if Z <
3	**Calculate Z Score**	Z_d = (x■ − μ) = 1.5 / (3.9/6) =
4	**Compare & Conclude**	Z_d = 2.31 > 1 p-value = 2 × (0.0208 < 0.05

Decision Boundary — Problem 1 (Two-Tail, 95% CI)



FINAL CONCLUSION: Z = 2.31 > 1.96 and p = 0.0208 < 0.05
We **REJECT the Null Hypothesis**. The average height is NOT 168 cm.
The sample suggests the average height appears to be increasing.

PROBLEM 2 — Bulb Warranty Example (Left-Tail Test)

A factory makes bulbs with an average warranty of **5 years** (σ = 0.50). A worker believes bulbs malfunction in **less than 5 years**. He tests **40 bulbs** and finds average life = **4.8 years**. At **2% significance level**, is there enough evidence?

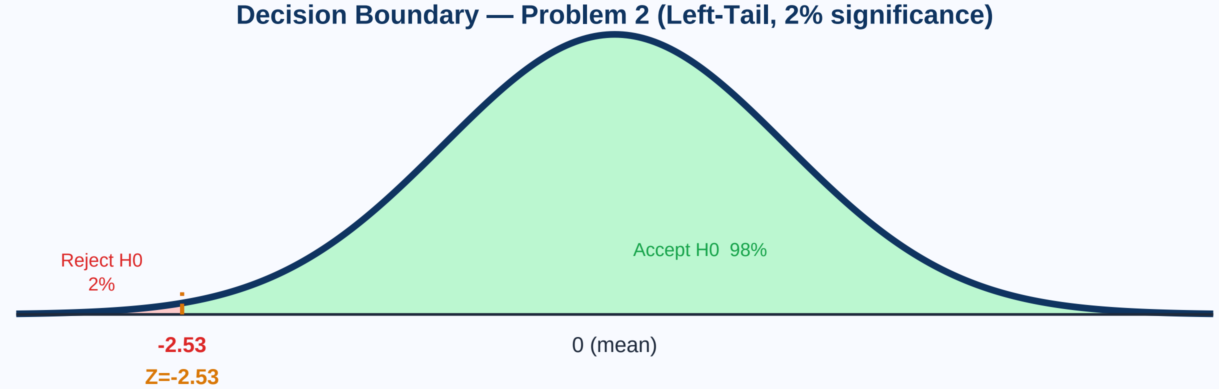
Given Values

μ	σ	n	x■	α (sig. level)	Tail
5 years	0.50	40	4.8 years	0.02 (2%)	LEFT (one-tail)

Step-by-Step Solution

1	**State Hypotheses**	H0: μ = 5 (warr H1: μ < 5 (warr
2	**Find Critical Z Value**	α = 0.02 → left From Z-table: a Reject H0 if Z <
3	**Calculate Z Score**	Z_d = (x■ − μ) = −0.2 / (0.50/6
4	**Find p-value**	Look up Z = −2 Note: 0.0570 in p-value = 0.005
5	**Compare & Conclude**	Compare p-val 0.0057 < 0.02 - Note: Notes inc

Decision Boundary — Problem 2 (Left-Tail, 2% significance)



NOTE ON p-VALUE ERROR IN NOTES:

$Z = -2.53 \rightarrow$ Z-table area = **0.0057** (NOT 0.0570 — decimal error in notes)

$0.0057 < 0.02$ (α) $\rightarrow$ TRUE $\rightarrow$ Correct conclusion is **REJECT H0**

CORRECT FINAL CONCLUSION: $Z = -2.53$ and $p = 0.0057 < 0.02$

We **REJECT the Null Hypothesis**. There IS evidence the bulb warranty should be revised.

QUICK REFERENCE CHEAT SHEET

How to Read the Z-Table

Situation	How to get p-value	Example
LEFT-TAIL (Z is negative, H1: μ < X)	p = table value directly	Z=–2.53 → p=0.0057
RIGHT-TAIL (Z is positive, H1: μ > X)	p = 1 – table value	Z=+2.31 → p=1–0.9896=0.0104
TWO-TAIL (H1: μ ≠ X)	p = 2 × (smaller tail area)	Z=2.31 → p=2×0.0104=0.0208

The Complete Z-Test Process

STEP 1	**Identify Given Values** μ, σ, n, x■, CI or α
STEP 2	**Set Hypotheses** H0 = claim, H1 = what you want to prove
STEP 3	**Determine Test Type** H1 uses ≠ → Two-Tail H1 uses < → Left-Tail H1 uses > → Right-Tail
STEP 4	**Find Critical Z** Look up Z-table using α/2 (two-tail) or α (one-tail)
STEP 5	**Calculate Z_d** Z_d = (x■ – μ) / (σ/√n)
STEP 6	**Find p-value** Use Z-table with your Z_d value
STEP 7	**Make Decision** p < α → Reject H0 p ≥ α → Fail to Reject H0

Both Problems Side by Side

	Problem 1 (Height)	Problem 2 (Bulb)
Test Type	Two-Tail	Left-Tail (One-Tail)
H0	μ = 168	μ = 5
H1	μ ≠ 168	μ < 5
α	0.05	0.02
Critical Z	±1.96	–2.53
Z_d	2.31	–2.53

p-value	0.0208	0.0057
Decision	REJECT H0 ✓	REJECT H0 ✓

Remember: $p < \alpha \rightarrow$ Reject H0 | Z-table always gives AREA TO THE LEFT | $n > 30$ + known $\sigma \rightarrow$ use Z-test